

Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools

A Systematic Synthesis of Empirical Evidence on Fabricated, Misattributed, and Corrupted References in AI-Generated Text

Abstract

Large language models (LLMs) and the agentic search systems built on top of them are increasingly used to locate, summarize, and cite scholarly and journalistic sources, yet a growing body of empirical work shows that these systems routinely generate citations to sources that do not exist or misattribute claims to real sources that do not support them. This paper synthesizes peer-reviewed and preprint evidence on citation fabrication rates across general-purpose chatbots (ChatGPT, Bard/Gemini, Claude, Copilot, Grok, DeepSeek), retrieval-augmented generation (RAG) pipelines, and agentic generative search engines (Bing Chat, Perplexity, NeevaAI, YouChat). Reported fabrication rates in the literature range from under 15% for retrieval-grounded, domain-specialized systems to over 90% for closed-book prompting of early-generation models, with substantial variation by discipline, prompt specificity, topic familiarity, and evaluation protocol. We organize this literature into a taxonomy distinguishing outright fabrication, misattribution, and metadata corruption, and we compare detection and mitigation approaches, including semantic entropy, sampling-based consistency methods, and dedicated citation-evaluation benchmarks such as ALCE, RAGTruth, and LegalCiteBench. We find that retrieval augmentation substantially reduces, but does not eliminate, fabrication; that agentic search engines with visible inline citations are not meaningfully more reliable than closed-book chatbots; and that no standardized, cross-model, cross-domain benchmark yet tracks citation fabrication longitudinally. We conclude by identifying methodological gaps — inconsistent operational definitions of “fabrication,” restricted auditability of proprietary agentic systems, sparse coverage of non-English and low-resource domains, and rapid model turnover that outdates point-in-time benchmarks — and propose a research agenda for standardized, reproducible citation-fabrication benchmarking.

Keywords: citation fabrication; hallucination; large language models; retrieval-augmented generation; agentic search engines; benchmarking; verifiability; academic integrity; legal artificial intelligence; misinformation

1. Introduction

Since the public release of ChatGPT in late 2022, large language models have moved rapidly from novelty chatbots to embedded research assistants, search agents, and drafting tools used across law, medicine, journalism, and the academy (Huang et al., 2025). A defining and persistent failure mode of these systems is “hallucination” — the generation of fluent, plausible content that is not grounded in fact (Ji et al., 2023; Huang et al., 2025). Within this broader category, citation fabrication is a particularly consequential subtype: LLMs and the search tools built on them frequently invent bibliographic references, case citations, or news attributions that appear stylistically authentic — correct author-name formats, plausible journal titles, well-formed digital object identifiers (DOIs) — but that do not correspond to any real publication, or that misrepresent the content of a publication that does exist (Walters & Wilder, 2023; Athaluri et al., 2023).

The practical stakes of citation fabrication are not hypothetical. Attorneys have been sanctioned by courts for submitting briefs containing fictitious case citations generated by ChatGPT, and commercial legal research vendors have advertised “hallucination-free” retrieval-augmented tools that independent audits subsequently found to hallucinate in 17–33% of queries (Magesh et al., 2025); multiple studies of ChatGPT- and Bard-generated literature reviews have found that a majority of cited references were entirely fabricated (Chelli et al., 2024). As students, researchers, journalists, clinicians, and legal professionals increasingly rely on LLM-based tools — including “agentic” search systems that browse the live web and append inline citations to their outputs — to locate and summarize sources, the reliability of machine-generated citations has become a question of direct relevance to information integrity, professional liability, and the scholarly record itself (Naser, 2026).

Despite the volume of individual studies documenting citation fabrication in specific models, domains, or tools, the literature remains fragmented: studies differ in their definitions of “fabrication” versus “inaccuracy,” in prompting protocols, in the models tested, and in whether systems have access to retrieval or live search. This fragmentation makes it difficult to answer basic comparative questions such as: Do agentic search tools with visible citations actually fabricate less than closed-book chatbots? Has model capability scaling reduced fabrication rates across successive generations? Does retrieval augmentation solve the problem, or merely relocate it?

This paper addresses these questions by systematically reviewing and benchmarking reported citation fabrication rates across the empirical literature on LLMs and agentic search tools. We make three contributions. First, we synthesize more than twenty independent empirical studies spanning medicine, law, the social sciences, humanities, economics, and general web search, and compile their reported fabrication and hallucination rates into a comparative framework. Second, we propose a taxonomy that distinguishes distinct failure modes — outright fabrication, misattribution, and metadata corruption — that are frequently conflated under the single label “hallucination.” Third, we critically assess current detection and mitigation approaches, including uncertainty-based detection methods, retrieval grounding, and dedicated citation-evaluation benchmarks, and we identify the methodological gaps that currently prevent standardized, longitudinal tracking of citation reliability across the LLM ecosystem.

The remainder of the paper is organized as follows. Section 2 reviews related work on hallucination in natural language generation broadly and on citation fabrication specifically. Section 3 presents a technical discussion organized around a taxonomy of fabrication types and a comparative synthesis of reported rates. Section 4 surveys current detection, mitigation, and benchmarking approaches. Section 5 discusses methodological challenges and limitations of the existing literature. Section 6 identifies research gaps and proposes directions for future work. Section 7 concludes.

2. Background, Related Work, and Literature Review

2.1 Hallucination in Natural Language Generation and Large Language Models

Hallucination — text that is fluent and syntactically coherent but factually unsupported — predates the LLM era and was first studied systematically in narrower natural language generation (NLG) tasks such as abstractive summarization, data-to-text generation, and machine translation (Ji et al., 2023). Ji et al.’s (2023) survey in ACM Computing Surveys established an early taxonomy distinguishing “intrinsic” hallucinations, which contradict source content, from “extrinsic” hallucinations, which introduce unverifiable information — a distinction later work extended to general-purpose LLMs. Huang et al. (2025), in a subsequent large-scale survey, refined this taxonomy for LLMs specifically, distinguishing factuality hallucinations (inconsistency with verifiable real-world facts) from faithfulness hallucinations (inconsistency with user instructions or supplied context), and cataloged causes spanning training-data noise, exposure bias during autoregressive decoding, and knowledge-boundary limitations.

A parallel line of work has focused on detecting hallucination without ground-truth labels. Manakul, Liusie, and Gales (2023) introduced SelfCheckGPT, a zero-resource, black-box method premised on the idea that sampled responses to the same query remain consistent when a model possesses genuine underlying knowledge but diverge stochastically when the model is confabulating. Farquhar, Kossen, Kuhn, and Gal (2024), publishing in *Nature*, extended this consistency-based intuition into “semantic entropy,” which clusters sampled generations by meaning rather than surface form before computing an entropy score, enabling unsupervised detection of “confabulations” — a narrower class of hallucination corresponding to arbitrary, incorrect generations driven by an absence of underlying knowledge. Both approaches demonstrate that hallucination, including citation fabrication, is at least partially predictable from the internal consistency of a model’s own sampling distribution, a property later exploited in citation-specific detection pipelines (Naser, 2026).

2.2 Citation Fabrication in General-Purpose Chatbots

The first systematic studies of citation fabrication emerged within months of ChatGPT’s public release. Athaluri et al. (2023) documented that ChatGPT frequently generated fake, peer-review-styled citations when asked to support scientific claims. Bhattacharyya, Miller, Bhattacharyya, and Miller (2023) prompted ChatGPT-3.5 to generate 30 short medical papers and found that a majority of the resulting references were either fabricated outright or contained

substantial inaccuracies across author, title, journal, year, volume, page, and PubMed identifier fields, with incorrect PubMed identifiers appearing in 93% of papers.

The most widely cited study in this literature is Walters and Wilder (2023), who prompted both ChatGPT-3.5 and ChatGPT-4 to generate short literature reviews across 42 multidisciplinary topics and manually verified all 636 resulting bibliographic citations. They found that 55% of GPT-3.5's citations and 18% of GPT-4's citations were entirely fabricated, and that even among non-fabricated citations, 43% (GPT-3.5) and 24% (GPT-4) contained substantive errors such as incorrect volume, issue, or page numbers. Fabricated citations disproportionately used real, plausible journal names paired with invented article titles, a pattern that complicates detection through surface-level plausibility checks alone.

Later work extended this basic paradigm across disciplines, models, and prompting conditions. Mugaanyi, Cai, Cheng, Lu, and Huang (2024) compared ChatGPT-3.5-generated citations across the natural sciences and humanities and found that DOI hallucination — the generation of a DOI that does not resolve to the cited work — occurred in 61.8% of natural-science citations versus 89.4% of humanities citations, attributing the disparity to the less structured, DOI-sparse character of humanities publishing. Buchanan, Hill, and Shapoval (2024) documented a comparable pattern in economics, finding that ChatGPT frequently attributed real claims to nonexistent working papers. At much larger scale, Naser (2026) audited ten commercially deployed LLMs across four academic domains, generating and verifying 69,557 citation instances against three scholarly databases (CrossRef, OpenAlex, and Semantic Scholar); hallucination rates spanned a fivefold range, from 11.4% to 56.8%, driven jointly by model identity, domain, and prompt framing, and the study found fabrication to be consistently prompt-induced rather than spontaneous — no model generated invented citations when not explicitly asked to cite sources.

Cross-model comparative studies conducted after the proliferation of competing chatbot products (Claude, Gemini, Copilot, Grok, DeepSeek, Perplexity, Le Chat) found substantial heterogeneity in reliability. Cabezas-Clavijo and Sidorenko-Bautista (2025) evaluated eight chatbots' free-tier bibliographic reference generation across five academic domains and 400 references, finding that only 26.5% of references were fully correct while 39.8% were erroneous or entirely fabricated; Grok and DeepSeek produced no outright fabrications in their sample, while Copilot, Perplexity, and Claude exhibited the highest fabrication rates — a finding notable because Copilot and Perplexity are agentic, web-connected search products rather than purely closed-book chatbots, which undercuts any assumption that live retrieval alone guarantees citation reliability.

2.3 Domain-Specific Studies: Medicine, Mental Health, and Law

Medical and mental-health applications have received particular scrutiny given the direct patient-safety implications of fabricated clinical evidence. Chelli et al. (2024) tested ChatGPT-3.5, ChatGPT-4, and Bard on their ability to replicate 11 published systematic reviews across four medical fields, finding hallucination rates of 39.6% for GPT-3.5, 28.6% for GPT-4, and 91.4% for Bard, with GPT-4 achieving only 13.4% precision in retrieving relevant papers. Linardon, Jarman, McClure, Anderson, Liu, and Messer (2025) extended this paradigm to test whether citation fabrication in GPT-4o varies systematically with topic familiarity, generating literature reviews on mental-health disorders that differ in public visibility and research maturity; fabrication was consistently higher for less publicly familiar, less-studied conditions than for widely studied conditions, suggesting that models fabricate more readily when their training data for a topic is sparse. Wu et al. (2025), publishing in *Nature Communications*, developed an automated framework for auditing whether LLM-generated medical citations actually support the specific claims attached to them — a distinction between a citation's existence and its relevance that recurs throughout this literature.

Legal applications have produced some of the most rigorously designed studies in this space, in part because the discrete, verifiable nature of case citations lends itself to systematic auditing. Dahl, Magesh, Suzgun, and Ho (2024) issued a battery of verifiable, closed-book legal queries about federal case law to ChatGPT-4 and other public models and found hallucination rates of at least 58%, rising to 88% for Llama 2; models also struggled to predict their own errors and frequently accepted users' factually incorrect legal premises without correction. Building on this baseline, Magesh, Surani, Dahl, Suzgun, Manning, and Ho (2025) conducted the first systematic, preregistered evaluation of commercial retrieval-augmented legal research tools — Lexis+ AI, Thomson Reuters' Ask Practical Law AI, and Westlaw AI-Assisted Research — each of which had been marketed by its vendor as "hallucination-free" or as eliminating or

avoiding hallucination. The audit found hallucination rates of 17% to 33% across the three commercial RAG-based tools, compared with 43% for a general-purpose GPT-4 baseline without retrieval — evidence that retrieval augmentation substantially reduces, but does not eliminate, citation fabrication, and that vendor marketing claims of “hallucination-free” performance were not empirically supported.

2.4 Verifiability and Citation Fabrication in Agentic Search Engines

A distinct strand of research evaluates “agentic” or generative search engines — systems such as Bing Chat, Perplexity, NeevaAI, and YouChat that combine live web retrieval with generated, inline-cited answers — using the concept of “verifiability”: responses should be both comprehensive (every factual statement is supported by a citation) and accurate (every citation actually supports its associated statement) (Liu, Zhang, & Liang, 2023). In the first large-scale human evaluation of this kind, Liu et al. (2023) audited four generative search engines and found that only 51.5% of generated sentences were fully supported by their citations and only 74.5% of citations accurately supported their associated statement — meaning roughly one in four inline citations displayed to users did not actually verify the claim it was attached to, despite the reassuring visual presence of a citation marker.

More recently, journalistic audits have extended this verifiability paradigm to news attribution specifically. Jaźwińska and Chandrasekar (2025), reporting for the Tow Center for Digital Journalism at Columbia Journalism Review, tested eight AI search tools (including ChatGPT, Perplexity, Copilot, Gemini, Grok, and DeepSeek) on their ability to correctly identify and attribute the source of specific news content, finding that most tools answered confidently even when they lacked accurate information rather than declining to answer; ChatGPT, for example, incorrectly identified the source of 134 of 200 tested articles while signaling uncertainty in only 15 of its 200 responses. This confident-but-wrong pattern — model outputs presented with the same assertive register regardless of underlying accuracy — recurs across the citation-fabrication literature and is a central concern for downstream users who cannot easily distinguish reliable from fabricated citations based on surface presentation alone.

3. Main Thematic and Technical Discussion

3.1 A Taxonomy of Citation Fabrication

Building on the distinctions introduced across the literature reviewed above, we propose a three-part taxonomy that separates failure modes frequently conflated under the general heading of “citation hallucination”:

Outright fabrication: the model generates a complete bibliographic reference, case citation, or source attribution corresponding to no real publication or document. Fabricated references in this category are stylistically well-formed — plausible author names, journal titles, and even syntactically valid DOIs — but resolve to nothing when checked against a bibliographic database (Walters & Wilder, 2023; Dahl et al., 2024).

Misattribution: the cited source exists and is a real publication, but it does not support, or actively contradicts, the specific claim to which it is attached. This is the failure mode targeted by “verifiability” audits of agentic search engines, where a citation marker is present and clickable but the underlying source does not substantiate the adjoining sentence (Liu et al., 2023; Wu et al., 2025).

Metadata corruption: the underlying work is real and is correctly identified as the intended source, but one or more bibliographic elements — author list, year, volume, page numbers, or DOI — is incorrect. This category, while less severe than outright fabrication, still frustrates verification and citation tracing, and is empirically common even among LLM citations classified as “real” (Walters & Wilder, 2023; Bhattacharyya et al., 2023; Mugaanyi et al., 2024).

These categories are not mutually exclusive in practice — a single LLM-generated reference list routinely contains instances of all three — but distinguishing them matters for both measurement and mitigation. Outright fabrication is detectable through database lookup alone; misattribution requires semantic comparison between a citation’s content and the sentence it supports, a substantially harder task that most large-scale audits acknowledge they measure only approximately; and metadata corruption, while easiest to detect, is often undercounted in studies that classify a reference as merely “existing” without checking every field.

3.2 Comparative Synthesis of Reported Rates

Table 1 synthesizes reported fabrication and hallucination rates across the studies reviewed in Section 2, spanning general-purpose chatbots, commercial legal research tools, and agentic generative search engines.

Study	System(s) Tested	Domain / Task	Reported Fabrication / Hallucination Rate
Walters & Wilder (2023)	ChatGPT-3.5; ChatGPT-4	Multidisciplinary literature reviews (closed-book)	55% fabricated (GPT-3.5); 18% fabricated (GPT-4); 43%/24% of remaining citations contained substantive errors
Bhattacharyya et al. (2023)	ChatGPT-3.5	Short medical papers (30 papers)	Majority of references fabricated or inaccurate; incorrect PMID in 93% of papers
Mugaanyi et al. (2024)	ChatGPT-3.5	Natural sciences vs. humanities citations	DOI hallucination: 61.8% (sciences) vs. 89.4% (humanities)
Chelli et al. (2024)	ChatGPT-3.5; ChatGPT-4; Bard	Systematic review replication (4 medical fields)	39.6% (GPT-3.5); 28.6% (GPT-4); 91.4% (Bard)
Linardon et al. (2025)	GPT-4o	Mental health literature reviews	Fabrication substantially higher for low- vs. high-familiarity topics
Dahl et al. (2024)	GPT-4; GPT-3.5; PaLM 2; Llama 2	U.S. federal case law (closed-book)	58%–88% hallucination, varying by model
Magesh et al. (2025)	Lexis+ AI; Ask Practical Law AI; Westlaw AI-AR; GPT-4 (baseline)	Commercial legal RAG research tools	17%–33% (RAG tools); 43% (non-retrieval GPT-4 baseline)
Liu, Zhang & Liang (2023)	Bing Chat; NeevaAI; Perplexity; YouChat	Open-domain generative web search	48.5% of statements unsupported by citation; 25.5% of citations inaccurate
Jaźwińska & Chandrasekar (2025)	8 AI search tools (ChatGPT, Perplexity, Copilot, Gemini, Grok, DeepSeek, etc.)	News source and quote attribution	Majority of tested attributions incorrect; low expressed uncertainty despite frequent errors
Cabezas-Clavijo & Sidorenko-Bautista (2025)	8 chatbots (ChatGPT, Claude, Copilot, DeepSeek, Gemini, Grok, Le Chat, Perplexity)	Academic bibliographic references (5 domains)	39.8% erroneous or fabricated overall; Copilot, Perplexity, Claude highest; Grok, DeepSeek lowest
Naser (2026)	10 commercial LLMs	Academic citations, 4 domains (69,557 instances)	11.4%–56.8% hallucination, varying by model, domain, and prompt framing
Gao et al. (2023) [ALCE]	GPT-3.5-class models with retrieval	Open QA with citations (ELI5, ASQA, QAMPARI)	≥50% of ELI5 responses lack complete citation support

Table 1. Reported citation fabrication and hallucination rates across representative studies, systems, and domains. Figures are drawn directly from each cited study and are not independently re-verified; definitions of “fabrication” and “hallucination” vary across studies (see Section 5).

The synthesis in Table 1 reveals several consistent patterns despite substantial heterogeneity in methodology. First, model generation matters: within the GPT family, GPT-4 consistently fabricates at roughly one-third to one-half the rate of GPT-3.5 across studies that test both (Walters & Wilder, 2023; Chelli et al., 2024), though Naser’s (2026) finding that improvements are not guaranteed when deploying newer LLMs across the wider multi-vendor landscape cautions against assuming that capability scaling monotonically reduces fabrication for every model family or provider.

Second, retrieval augmentation reduces but does not eliminate fabrication. The clearest evidence comes from the legal domain, where Magesh et al. (2025) found that commercial RAG-based legal tools reduced hallucination roughly by

half relative to a non-retrieval GPT-4 baseline (17–33% versus 43%) but did not approach the “hallucination-free” performance advertised by vendors. This pattern — meaningful but partial improvement from grounding — recurs in the RAGTruth benchmark’s finding that retrieval-augmented systems still hallucinate detectably across question answering, summarization, and data-to-text tasks even when the correct source document is present in context (Niu et al., 2024).

Third, and counterintuitively, the presence of an agentic, web-connected retrieval capability and a visible citation interface does not reliably indicate higher accuracy. Liu et al.’s (2023) verifiability audit of Bing Chat, NeevaAI, Perplexity, and YouChat found citation-support rates in a similar range to closed-book chatbot fabrication rates reported elsewhere, and Cabezas-Clavijo and Sidorenko-Bautista (2025) found that two web-connected products, Copilot and Perplexity, exhibited among the highest fabrication rates of the eight chatbots tested, alongside Claude. This suggests that visible citation markers can create a false impression of verifiability — users may extend more trust to an inline-cited answer than to an unadorned chatbot response, even though the underlying accuracy of the citation is not guaranteed by the presence of the citation interface element itself (Jaźwińska & Chandrasekar, 2025).

Fourth, domain and topic familiarity are strong moderators. Fabrication is consistently higher in less-structured, DOI-sparse literatures such as the humanities relative to the natural sciences (Mugaanyi et al., 2024), and higher for less publicly visible topics within a single field (Linardon et al., 2025). Naser (2026) further found that fabrication is prompt-induced rather than spontaneous, and that popularity-biased retrieval — models preferentially citing highly cited, prominent works — compounds visibility disparities in what gets cited accurately versus fabricated.

4. Current Approaches and Developments

4.1 Grounding via Retrieval-Augmented Generation

The dominant industry response to citation fabrication has been retrieval-augmented generation (RAG), in which a model’s output is conditioned on retrieved source documents rather than generated purely from parametric memory. The evidentiary record on RAG’s effectiveness is mixed but broadly positive: Magesh et al. (2025) documented meaningful reductions in legal hallucination rates for RAG-based commercial tools relative to non-grounded baselines, and industry vendors including Lexis and Thomson Reuters have marketed RAG as effectively solving the hallucination problem. However, the same study’s central finding was that these marketing claims of “hallucination-free” performance are not supported empirically — RAG-based systems still hallucinated in 17–33% of tested queries, and errors included both outright fabrication of case law and more subtle mischaracterization of retrieved authorities. The RAGTruth benchmark (Niu et al., 2024) independently corroborates this pattern at a larger, model-agnostic scale, providing nearly 18,000 human-annotated examples spanning question answering, summarization, and data-to-text generation with both response- and span-level hallucination labels, and demonstrating that hallucination persists across model families even when correct grounding documents are supplied in context.

4.2 Detection Methods

Where fabrication cannot be entirely prevented at generation time, a parallel research direction has focused on post hoc or inference-time detection. Sampling-based consistency methods, exemplified by SelfCheckGPT (Manakul et al., 2023), detect likely hallucination by measuring the divergence among multiple sampled responses to the same prompt, on the premise that genuine knowledge produces convergent answers while confabulation produces divergent ones. Semantic entropy (Farquhar et al., 2024) refines this intuition by clustering sampled generations according to semantic equivalence — assessed via natural language inference — before computing an entropy measure over the resulting clusters, allowing the method to distinguish cases where a model expresses the same correct fact in varied phrasing from cases where sampled outputs genuinely disagree in meaning. Naser (2026) adapted this consistency logic directly to citation verification, showing that multi-model consensus — citations independently produced by more than three of ten tested LLMs — achieved 95.6% verified accuracy, a 5.8-fold improvement over single-model citation reliability, and that within-prompt repetition (requiring more than two consistent repetitions of the same citation) achieved 88.9% accuracy, offering a practical, model-agnostic filter that end users or downstream tools could apply without access to any single vendor’s internal uncertainty signals.

4.3 Dedicated Citation and Verifiability Benchmarks

A growing set of benchmarks specifically targets citation quality rather than general factuality. ALCE (Gao, Yen, Yu, & Chen, 2023) was among the first benchmarks purpose-built for evaluating LLM-generated text with citations, combining fluency, correctness, and citation-quality metrics across three question-answering datasets; the study found that even the strongest contemporary models left roughly half of ELI5 responses without complete citation support, establishing citation quality as a distinct and non-trivial capability separate from raw factual accuracy. Liu et al.'s (2023) verifiability framework, applying comparable recall and precision logic to commercial generative search engines rather than research prototypes, remains a primary reference point for evaluating deployed agentic search products rather than research systems.

Domain-specific benchmarks have since proliferated. LegalCiteBench (Chen, Yin, & Zhou, 2026) targets the legal domain specifically, evaluating closed-book citation recovery, citation completion, error detection, and case matching across roughly 24,000 evaluation instances derived from real U.S. case law, directly operationalizing the typology of legal hallucination introduced by Dahl et al. (2024) into a reusable benchmark suite. In parallel, hallucination-detection benchmarks not specific to citations, such as RAGTruth (Niu et al., 2024), provide the underlying annotated data on which citation-specific tools and detectors are increasingly built and fine-tuned.

4.4 Practical Auditing of Deployed Systems

Alongside academic benchmark construction, journalistic and library-science audits have played a distinct and complementary role in evaluating deployed, black-box commercial systems that are not otherwise open to systematic benchmarking. The Tow Center's repeated audits of consumer AI search tools (Jaźwińska & Chandrasekar, 2025) exemplify this approach: rather than constructing a benchmark dataset with ground-truth labels in advance, these audits pose realistic, adversarially chosen queries — including queries about content that publishers have technically blocked from AI crawlers — and manually verify the resulting citations and attributions against the actual source content. This method captures deployment-specific behaviors, such as a system's tendency to answer confidently rather than decline when it lacks reliable information, that are not always visible in academic benchmarks built around a fixed, pre-selected query set.

5. Research Challenges and Limitations

Several methodological challenges limit the comparability and durability of the citation-fabrication literature reviewed above.

Inconsistent operational definitions. Studies vary in what counts as a “fabricated” versus “inaccurate” versus merely “incomplete” citation, and in whether metadata corruption is counted toward the headline hallucination rate at all. Chelli et al. (2024) and Walters and Wilder (2023), for example, both report separate fabrication and error rates but define “error” using different bibliographic-element checklists, complicating direct numerical comparison across studies. This inconsistency means that the reported rates synthesized in Section 3 should be read as broadly indicative of relative severity and directionality rather than as precisely comparable point estimates.

Prompt and protocol sensitivity. Fabrication rates are highly sensitive to prompting conditions — closed-book versus retrieval-enabled, general versus specialized prompts, and single- versus multi-turn interaction — and Naser (2026) found that citation fabrication is essentially prompt-induced, meaning headline rates reported by any single study describe that study's specific elicitation protocol rather than a fixed, model-intrinsic property. Linardon et al.'s (2025) finding that fabrication varies with topic familiarity, and Mugaanyi et al.'s (2024) finding that it varies with disciplinary structure, further indicate that any single reported rate is a function of the query distribution tested, not a universal constant for a given model.

Rapid model turnover and benchmark decay. The pace of LLM release cycles means that point-in-time benchmark results decay quickly in relevance. Several systems tested in early studies (Bard, early GPT-3.5 deployments) have since been retired or substantially retrained, and Naser's (2026) explicit finding that improvements are not guaranteed when deploying newer LLMs undercuts any simple assumption that citation reliability improves monotonically with each new release. Static, one-time benchmark papers are consequently poorly suited to tracking a rapidly moving target, and

the literature currently lacks any recurring, longitudinally maintained leaderboard analogous to general-capability leaderboards.

Restricted auditability of proprietary agentic systems. Unlike open-weight models or transparent research prototypes, commercial agentic search products and legal RAG tools are closed systems whose retrieval corpora, ranking logic, and grounding mechanisms are not disclosed. Magesh et al. (2025) note that evaluating proprietary legal AI tools required manually submitting hand-crafted queries through vendor web interfaces rather than programmatic, large-scale API access, a constraint that limits sample sizes and reproducibility relative to benchmarks built on open models. This opacity also means independent researchers cannot easily determine whether observed fabrication stems from retrieval failure, generation failure, or an interaction between the two.

Underrepresentation of non-English and low-resource domains. Nearly all of the studies synthesized in this paper test English-language queries and predominantly Western academic or legal corpora. Given the documented sensitivity of fabrication rates to training-data density and topic familiarity (Linardon et al., 2025; Mugaanyi et al., 2024), there is limited empirical basis for assuming that reported rates generalize to non-English literatures, non-U.S. legal systems, or citation practices in lower-resource research communities — a gap with direct equity implications given that these are plausibly the settings where fabrication risk is highest and independent verification capacity is lowest.

Measurement cost and scalability trade-offs. Rigorous verification of citation accuracy — particularly for the misattribution category, which requires semantic judgment of whether a real, correctly identified source actually supports a given claim — is labor-intensive and difficult to fully automate, forcing most published audits to rely on modest sample sizes verified by human annotators. Naser’s (2026) 69,557-citation audit is a notable exception, but it relies on database-lookup verification rather than the finer-grained semantic verification of whether a real citation actually supports its associated claim, meaning even the largest available audits cannot fully capture the misattribution category defined in Section 3.

6. Research Gaps and Future Directions

Based on the challenges identified above, we propose the following directions for future research.

Standardized, shared benchmark protocols. The field would benefit from a shared task analogous to ALCE (Gao et al., 2023) or RAGTruth (Niu et al., 2024) but purpose-built for citation fabrication specifically, with a fixed, versioned definition distinguishing outright fabrication, misattribution, and metadata corruption, applied consistently across models and domains. LegalCiteBench (Chen et al., 2026) offers a domain-specific template for this kind of standardization that could be extended to medicine, the social sciences, and general web search.

Longitudinal, recurring evaluation infrastructure. Given documented evidence that citation reliability does not improve monotonically across model releases (Naser, 2026), the field needs recurring, versioned audits — analogous to general-capability leaderboards — that re-test each major model release against a fixed query set over time, rather than relying on isolated, one-time snapshot studies that quickly become outdated.

Automated, scalable misattribution detection. Because outright fabrication is comparatively easy to detect via database lookup while misattribution requires semantic judgment, future work should prioritize scalable automated methods for verifying whether a real, correctly identified citation actually supports its associated claim — potentially building on natural-language-inference-based approaches already used in semantic entropy (Farquhar et al., 2024) and SelfCheckGPT (Manakul et al., 2023), but applied specifically to claim–citation pairs rather than whole-response consistency.

Cross-lingual and low-resource domain coverage. Future benchmarking efforts should deliberately extend beyond English-language, Western-centric corpora to test whether fabrication rates and their moderators (topic familiarity, disciplinary structure) generalize to other languages, legal systems, and research communities, given the equity implications of differential reliability documented within English-language studies alone (Mugaanyi et al., 2024).

Evaluating agentic tool-use chains, not just final outputs. As agentic systems increasingly involve multi-step tool use — issuing search queries, selecting among retrieved results, and synthesizing citations across multiple retrieval calls —

future work should evaluate fabrication risk at each stage of the agentic pipeline rather than solely at the level of the final generated citation, since errors may originate in query formulation or source selection rather than in the generation step alone.

Independent, standardized auditing of commercial claims. Given the documented gap between vendor marketing claims of “hallucination-free” performance and independently measured hallucination rates of 17–33% for commercial legal RAG tools (Magesh et al., 2025), there is a clear case for independent, standardized, and recurring auditing infrastructure — potentially housed in a neutral academic or journalistic institution — capable of evaluating proprietary commercial AI research and search tools on a comparable, ongoing basis, extending the model exemplified by the Tow Center’s news-citation audits (Jaźwińska & Chandrasekar, 2025) to legal, medical, and academic research tools.

7. Conclusion

The empirical literature reviewed in this paper converges on several robust findings despite methodological heterogeneity. Citation fabrication is a pervasive, well-documented failure mode across essentially every large language model and agentic search tool that has been independently audited, with reported rates ranging from roughly one in ten citations to more than nine in ten depending on model, domain, and prompting protocol. Retrieval augmentation meaningfully reduces fabrication relative to closed-book generation but does not eliminate it, contradicting vendor claims of “hallucination-free” performance in at least one rigorously audited commercial domain. Agentic search engines with visible inline citations are not inherently more reliable than ordinary chatbots, and in several studies exhibited comparable or higher fabrication rates than closed-book models, suggesting that the visual presence of a citation should not be mistaken for verified accuracy. Fabrication is systematically higher for less well-represented topics and disciplines — and, by extension, plausibly for non-English and low-resource domains not yet well studied.

Benchmarking citation fabrication rigorously and comparably across this rapidly evolving landscape remains an open problem. The field currently lacks standardized definitions, recurring longitudinal evaluation infrastructure, and independent auditing capacity commensurate with the scale at which these tools are now deployed in law, medicine, journalism, and scholarship. As LLM-based agentic tools become further embedded in professional and academic research workflows, closing this benchmarking gap is not merely an academic exercise but a precondition for responsible deployment: users of these systems currently have no reliable, up-to-date, cross-vendor basis for calibrating how much trust a machine-generated citation actually deserves.

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